

Entropy + Bit-Plane Fusion for AI-Generated Image Detection

Combining MLEP (NeurIPS 2025) and LOTA (ICCV 2025) into one trainable, dual-cue forensic classifier.

01 Problem Statement

Given an image, decide whether it is a real photograph or AI-generated (GAN/diffusion), and do so in a way that *generalizes* to generators the model never saw during training — the core failure mode of most existing detectors. Instead of learning content/semantics, this project deliberately reads low-level statistical fingerprints that generative models leave behind: unusual entropy structure and unusual bit-plane noise.

02 Source Papers & Code

MLEP — Multi-granularity Local Entropy Patterns

Yuan, Li, Zhang, Zhang, Li, Gao. **NeurIPS 2025**. Computes Shannon-entropy feature maps from shuffled small patches at multiple scales, then trains a standard CNN classifier on them. Trained on just 4 ProGAN object classes, evaluated across 32 distinct generative models for generalization.

Code: github.com/fkeufss/MLEP — official, training/testing scripts, pretrained checkpoint.

LOTA — Bit-Planes Guided Detection

Wang, Cheng, Zhang, Han, Gui. **ICCV 2025**, pp. 17246–17255. Decomposes images into 8 bit-planes, composes the least-significant bit-planes into a noise map, selects the highest-gradient patch, then classifies. Pretrained weights released for Stable Diffusion v1.4/v1.5.

Code: github.com/hongsong-wang/LOTA — official, ~14 stars (small but real, working repo with pretrained weights).

03 Dataset

GenImage (ProGAN subset + cross-generator test)

MLEP trains on 4 categories (car, cat, chair, horse) from ProGAN-generated images — a small, fast-to-download subset. Generalization is then tested against 32 different generative models, which is where the "does it really generalize" story comes from. LOTA additionally provides pretrained weights evaluated on Stable Diffusion v1.4/v1.5 outputs via GenImage.

04 Architecture — the fusion is your contribution

Branch A — Entropy

Patch shuffle → multi-scale resampling → local entropy pattern → CNN (ResNet-50)

Branch B — Bit-Plane

8 bit-plane decomposition → least-bit noise map → gradient-based patch selection → classifier head

Fusion Head (new)

Small gating/attention layer combining both branches' features into one real/fake decision

Neither source paper combines these two cues. Building the fusion head — and justifying why entropy and bit-plane noise are complementary rather than redundant — is the original piece of work you're contributing.

05 Does This Fulfill Both CV and DL?

COMPUTER VISION

DEEP LEARNING

✓ Operates directly on image pixels
— bit-plane decomposition and entropy computation are visual feature engineering

✓ Trains a real CNN classifier (ResNet-50) end-to-end, not just inference

✓ Patch-based visual analysis (gradient-driven patch selection, multi-scale resampling)

✓ Original fusion-head architecture design — the part neither paper did

✓ Extension work (Grad-CAM style localization) adds a genuine visual grounding task

✓ Loss design, optimizer choices, and hyperparameter ablations (patch size, masking ratio, resampling scales)

Honest caveat: the CV side here is *image forensics / low-level statistical vision* — it deliberately avoids semantic understanding (no object recognition, no scene understanding). This is a legitimate, established CV subfield, but it's a narrower slice of "computer vision" than something like segmentation or detection. Worth framing explicitly in your report so the examiner isn't expecting object-level visual reasoning.

✓ Computer Vision

✓ Deep Learning

Fulfills both halves of the brief. DL weight is strong (a real trainable architecture you designed). CV weight is real but narrow — pixel/statistical forensics rather than semantic vision — so name that scope explicitly rather than letting it be assumed.

06 Hardware Fit — Your Laptops

Windows • RTX 4050 (6GB) • i5-13th gen • 16GB RAM

ResNet-50 at batch 32 in mixed precision (fp16) comfortably fits in 6GB VRAM — this is the machine that does the actual training for both branches and the fusion head.

MacBook Air M4

Use for dataset preprocessing, running inference, and exporting the final fused model to Core ML/ONNX for a live webcam demo.

07 Extension Features to Build

- 1 Fusion head combining entropy + bit-plane branches — the core novel contribution
- 2 Explainability overlay: *where* the entropy/bit-plane anomaly concentrates visualize (Grad-CAM style)
- 3 Cross-generator stress test: generate your own held-out fakes and test generalization beyond GenImage's training distribution
- 4 Robustness study: test degradation under JPEG recompression / social-media resizing, propose a fix
- 5 Live demo: export to Core ML, run as a webcam classifier on the M4 Air

08 Suggested Timeline

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|--------|---|
| Wk 1–2 | Reproduce MLEP and LOTA separately on GenImage subset |
| Wk 3–4 | Design and train the fusion head |
| Wk 5–6 | Cross-generator generalization test with self-generated fakes |
| Wk 7 | Explainability overlay + robustness study |
| Wk 8 | Live demo, write-up, slides |

Plagiarism note: the fusion architecture, the cross-generator experiments, and all training/evaluation code are original work — only the two baseline papers' core ideas are

